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The transcription factor BCL11A restores differentiation potential to aged oligodendrocyte progenitor cells

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eLife Assessment

This **valuable** study demonstrates molecular changes associated with age related impairment in oligodendrocyte differentiation and ability to myelinate. The identification of particular genes that are associated with this decline will provide potential future targets for therapeutic interventions. The reviewers felt that the quality of the evidence was **solid** while identifying some minor weaknesses that if addressed would enhance the rigor of the study.

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Abstract

In young animals, oligodendrocyte progenitor cells (OPCs) undergo robust differentiation, progressing through stages to become pre-myelinating oligodendrocytes (Pre-OL) and ultimately myelinating oligodendrocytes (OLs). However, OPCs from aged animals have reduced differentiation ability. This disrupts myelin maintenance in the central nervous system (CNS), leading to a lack of remyelination following demyelinating injury and impaired adaptive myelination as a mechanism of learning. To uncover novel factors essential for restoring resilience in aged OPCs, we employed a data-driven approach involving the development of a computational pipeline, gSWITCH (accessible at: <https://altoslabs.shinyapps.io/gSWITCH/>), that allows for capture of precise dynamic gene expression patterns to pinpoint potential 'switch' genes during lineage progression. Using gSWITCH to identify potential switch genes crucial for OPCs, we conducted a comparative analysis of gene expression in OPCs isolated from young and aged animals. This revealed a group of transcription factors with decreased expression in aged OPCs. Further analysis of transcription factor binding site enrichment in OPC switches highlighted Bcl11a, a zinc finger transcription factor that could potentially serve as a master regulator of many switch genes. Ectopic overexpression of Bcl11a in aged OPCs did not enhance their proliferation; however, it significantly enhanced their differentiation into OLs. Overexpression of Bcl11a in aged mice, followed by demyelination injury in spinal cord white matter, significantly increased the differentiation of OPCs into OLs within the injury region compared to control aged mice. Furthermore, we found that Bcl11a is absent in invertebrates and has undergone pervasive purifying selection throughout vertebrate evolution, constraining its amino-acid sequence by eliminating deleterious mutations. Our study shows that reversing the age-related decline of this evolutionarily conserved factor in aged OPCs restores their impaired capacity for differentiation.

Introduction

The regulation of gene expression during development is highly dynamic, guiding the transition from progenitor cells to terminally differentiated cell type (Peter and Davidson, 2011 [↗](#); Pope and Medzhitov, 2018 [↗](#)). This temporal control ensures that appropriate gene expression programs are activated at each stage of lineage progression. Mapping these regulatory programs and identifying key driver within gene networks is essential for understanding how cell fate decisions are made and maintained. However, this developmental plasticity reduces with age, as progenitor cells progressively lose their ability to differentiate, impairing tissue homeostasis and repair (Rando et al., 2025 [↗](#); Brunet et al., 2023 [↗](#)).

In the central nervous system (CNS), oligodendrocytes (OLs) are responsible for producing myelin, the lipid-rich sheath that insulates axons and accelerates electrical impulse conduction (Nave and Werner, 2014 [↗](#)). In young animals, oligodendrocyte progenitor cells (OPCs) exhibit strong differentiation potential, transitioning through defined intermediate stages to generate mature, myelinating OLs. With ageing, however, this capacity declines: OPCs in older individuals show reduced ability to complete lineage progression, leading to compromised myelin maintenance, poor remyelination following demyelination injury and impaired adaptive myelination as a mechanism of learning (Franklin and French-Constant, 2017 [↗](#)). The molecular mechanisms underlying this age-related block in OPC differentiation remain poorly understood. Elucidating the regulatory factors that control OL lineage progression—and identifying their mis-regulation in aged OPCs—could uncover novel targets to restore resilience in these cells.

Here, we employed a data-driven approach to uncover novel factors essential for restoring resilience in aged OPCs. This involved the development of a computational pipeline, gSWITCH (accessible at: <https://altoslabs.shinyapps.io/gSWITCH/> [↗](#)), designed to capture precise dynamic gene expression patterns and identify potential ‘switch’ genes, including transcription factors, during lineage progression. Using this set of candidate switch genes, we performed a comparative analysis of gene expression between young and aged animals, which revealed a group of transcription factors downregulated in aged OPCs. Further analyses highlighted Bcl11a as a potential master regulator. Bcl11a expression was highest in OPCs and progressively decreased in Pre-OLs and OLs. Functional assays demonstrated that Bcl11a inhibition markedly reduced OPC differentiation into OLs. Overexpression of Bcl11a significantly enhanced differentiation in aged OPCs *in vitro* and following demyelination injury in the spinal cord of aged mice, without increasing OPC proliferation. Bcl11a is found only in vertebrates, and our molecular evolution analysis suggests that purifying selection has pervasively acted on this gene to preserve its amino acid sequences throughout vertebrate evolution. Our study showed that reversing the age-associated decline in expression of this evolutionarily conserved factor reinstates the differentiation capacity of aged OPCs.

Results

Identification of Switch genes during OL lineage progression

We performed RNA-seq analysis on OPC, Pre-OL and OL isolated from young rat (2-3 months) brains (Figure 1A [↗](#)). The OPC, Pre-OL, and OL samples were clearly distinguishable based on their global gene expression profiles (Figure 1B [↗](#)).

To identify critical genes during oligodendrocyte lineage progression, we developed gSWITCH application (see Methods and Supplementary Figure 1 [↗](#)). Notably, this biologist-friendly tool can be applied to identify key genes in any variable-dependent or time-series experiment. We identified four distinct dynamic patterns of gene expression, termed Type 1 to Type 4 switches (Figure 1C [↗](#), Supplementary Table 1 [↗](#)), during OL-lineage progression. These patterns are defined by the relative expression levels of specific gene sets across different cellular states. Disruptions to these expression levels can impair the coordinated dynamics and hinder lineage progression.

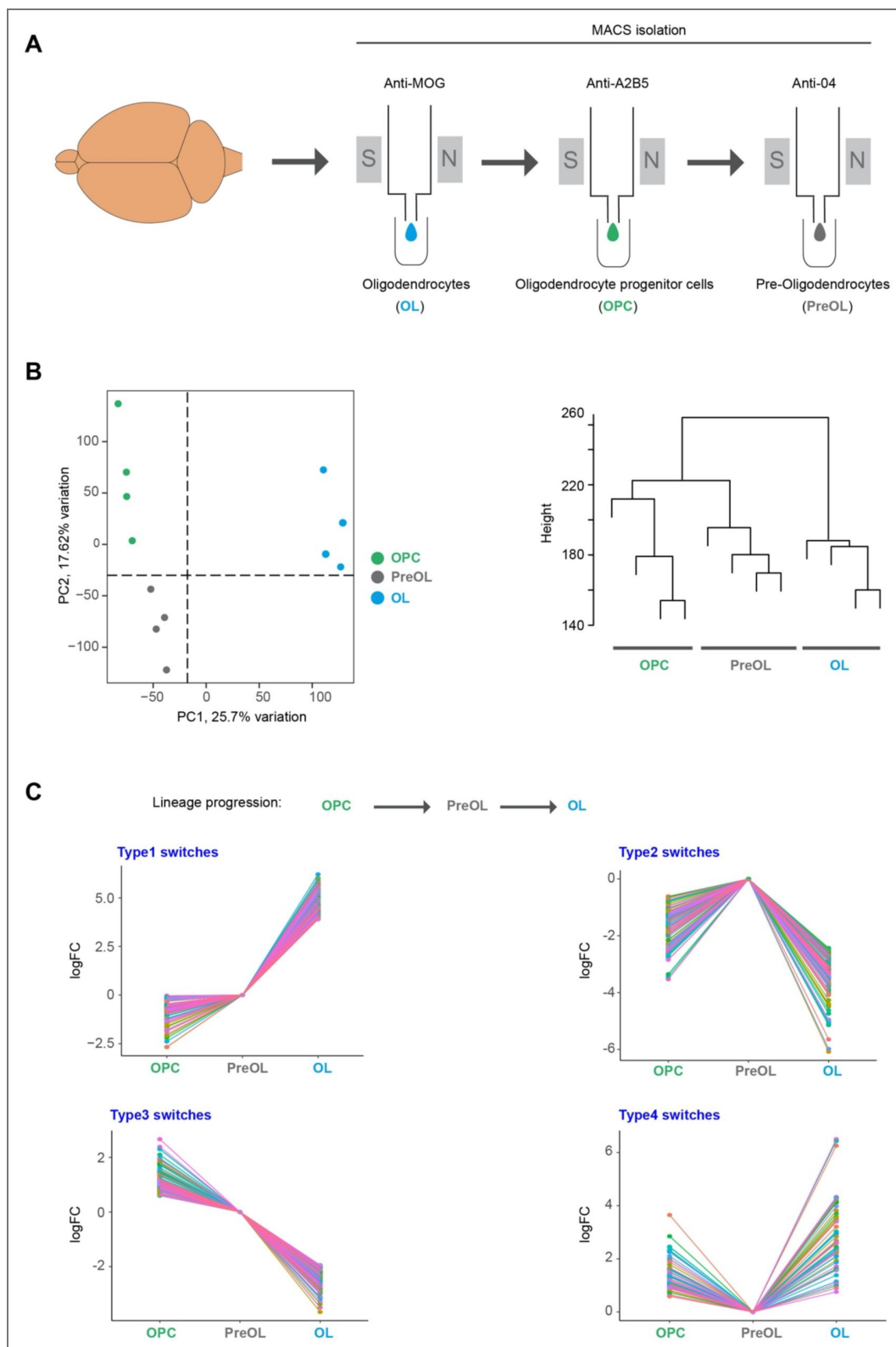


Figure 1. Dynamic regulation of genes during oligodendrocytes lineage progression.

A Schematic diagram showing magnetic-activated cell sorting (MACS)-based isolation of oligodendrocyte progenitor cells (OPCs), pre-oligodendrocytes (PreOLs), and oligodendrocytes (OLs) from young rat brains (2–3 months). The schema illustrates the order in which these cell types were most effectively isolated: mature OLs were isolated first using the OL-specific marker MOG, followed by OPCs using A2B5, and finally PreOLs using O4. **B** Principal component analysis (PCA) (left) and average-linkage hierarchical clustering (right) based on global gene expression show that each cell type forms a distinct cluster. OPCs, PreOLs, and OLs were isolated from four different rats. **C** Using gSWITCH, we identified four distinct patterns of switch genes during OL lineage progression. **(B–C)** GEO accession number of RNA-seq data: GSE301578.

Among the four expression patterns, Types 3 and 4 are most characteristic of OPCs, as they show higher relative expression in OPCs than in the next lineage stage, PreOL. For Type 4, expression levels rise again from PreOL to OL, but note that they do not return to the levels observed in OPCs. Indeed, gSWITCH is designed to detect a statistically significant difference between the first and last cell states (here, OPC vs OL), based on a user-defined fold-change threshold (1.5 in our case) (Supplementary Figure 1 [↗](#)).

Altered expression of switches in aged OPCs

We next examined the regulation of type 3 and type 4 switches in aged OPCs. Our transcriptome meta-analysis showed clear differences between aged and young OPCs, identifying a set of genes that are consistently upregulated or downregulated in aged OPCs (Figure 2A [↗](#), Figure 2B [↗](#)). A significant proportion of type 3 and type 4 switches were among these differentially expressed genes (135 out of 418 type 3 genes; 8 out of 45 type 4 genes). Notably, these switches were significantly downregulated in aged OPCs (Figure 2C [↗](#)), suggesting that altered regulation of these critical genes may impair the differentiation of aged OPCs into OLs.

Bcl11a is identified as a potential master switch

The orchestrated action of transcription factors (TFs) gradually drives the developmental program (Spitz F and Furlong EE, 2012 [↗](#)). Using gSWITCH, we identified switches that are TFs and found 67 TFs among type 3 and type 4 switches (Supplementary Table 2 [↗](#)).

Because a single transcription factor (TF) can regulate the expression of many genes, we aimed to identify TFs that may regulate type 3 and type 4 switches. TF binding site enrichment analysis using two chromatin immunoprecipitation (ChIP)-sequencing experimental evidence-based databases, ENCODE and ReMap2, predicted 27 TFs common to both datasets (Figure 3A [↗](#)). We then examined whether any of these 27 TFs were present in type 3 and type 4 switches and identified five [Zinc finger protein B-cell CLL/Lymphoma 11A (Bcl11a) (also known as CTIP1 (COUP-TF-interacting protein 1), CCAAT/enhancer-binding protein beta (Cebpb), Forkhead box M1 (Foxm1), Jun Proto-Oncogene, AP-1 Transcription Factor Subunit (Jun), and PR Domain Zinc Finger Protein 1 (Prdm1)], all of which were defined as type 3 switches (Figure 3A [↗](#), Figure 3B [↗](#)). Among these, only Bcl11a and Foxm1 were significantly downregulated in aged OPCs (Figure 3B [↗](#)). We focused further on Bcl11a, as its magnitude of change being greater than that of Foxm1 (Figure 3B [↗](#)).

The expression of Bcl11a followed a typical type 3 pattern: it was highest in OPCs and gradually decreased during differentiation, reaching significantly lower levels in OLs (Figure 3C [↗](#)). We therefore experimentally tested whether Bcl11a plays a functional role in OPC-to-OL differentiation, and whether restoring its expression can rescue the impaired differentiation capacity of aged OPCs.

Inhibition of Bcl11a in OPCs markedly reduced their differentiation

Since Bcl11a expression decreases as OPCs differentiate into OLs (Figure 3C [↗](#)), we addressed the hypothesis that, if Bcl11a acts as a master switch for lineage progression, its early expression in OPCs would be functionally required for differentiation to proceed. To test this, we used small interfering RNA (siRNA)-mediated knockdown. Transfection with siRNA targeting Bcl11a (siBcl11a) reduced its expression by approximately 79% in proliferating OPCs (Figure 4A [↗](#)). Using the 5-ethynyl-2'-deoxyuridine (EdU) incorporation assay, we found no significant effect of Bcl11a knockdown on OPCs proliferation (Figure 4B [↗](#)). However, Bcl11a inhibition led to a substantial (~77%) reduction in differentiation, as measured by immunostaining for myelin basic protein (MBP), a marker of OL differentiation (Figure 4C [↗](#)). These findings suggest that as a type 3 switch Bcl11a function is essential for priming OPCs for efficient differentiation.

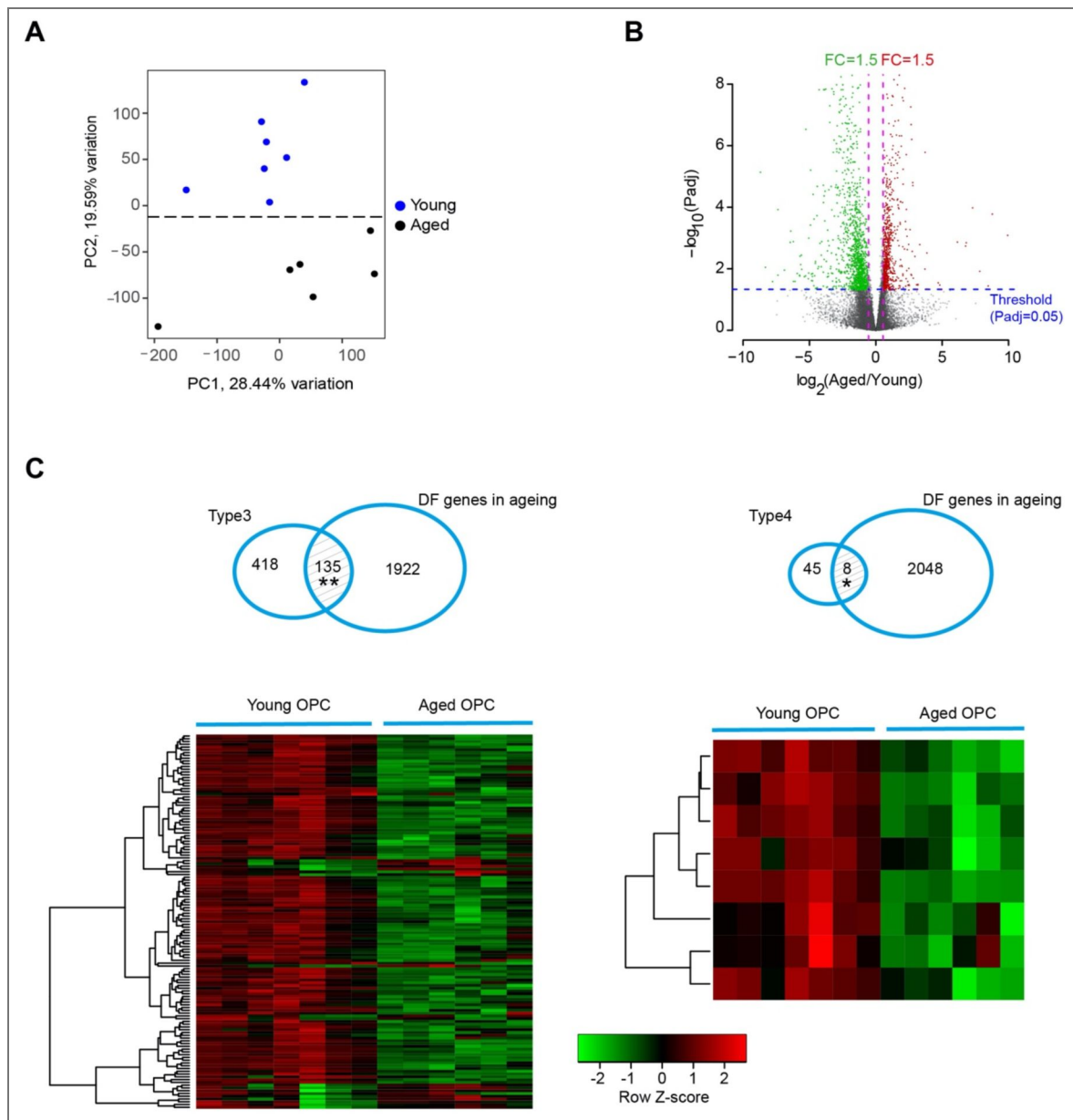


Figure 2. Down-regulation of switch genes due to ageing.

A) Principal component analysis (PCA) shows that aged and young OPCs are distinguishable based on global gene expression (meta-analysis of RNA-seq datasets, GEO accession number: GSE303317). **B**) Volcano plot showing differentially expressed genes in aged OPCs compared to young OPCs. $n = 6$ (aged rats), 7 (young rats), FC: fold change; Padj: adjusted p-value (Benjamini-Hochberg FDR after Wald test). **C**) Venn diagram (top) showing the subset of Type 3 and Type 4 switch genes affected by ageing. Their expression patterns are shown in the heatmap (bottom), where we observed down-regulation with ageing. * $P < 0.05$, ** $P < 0.001$, Chi-square test (two sided) with Yates' correction.

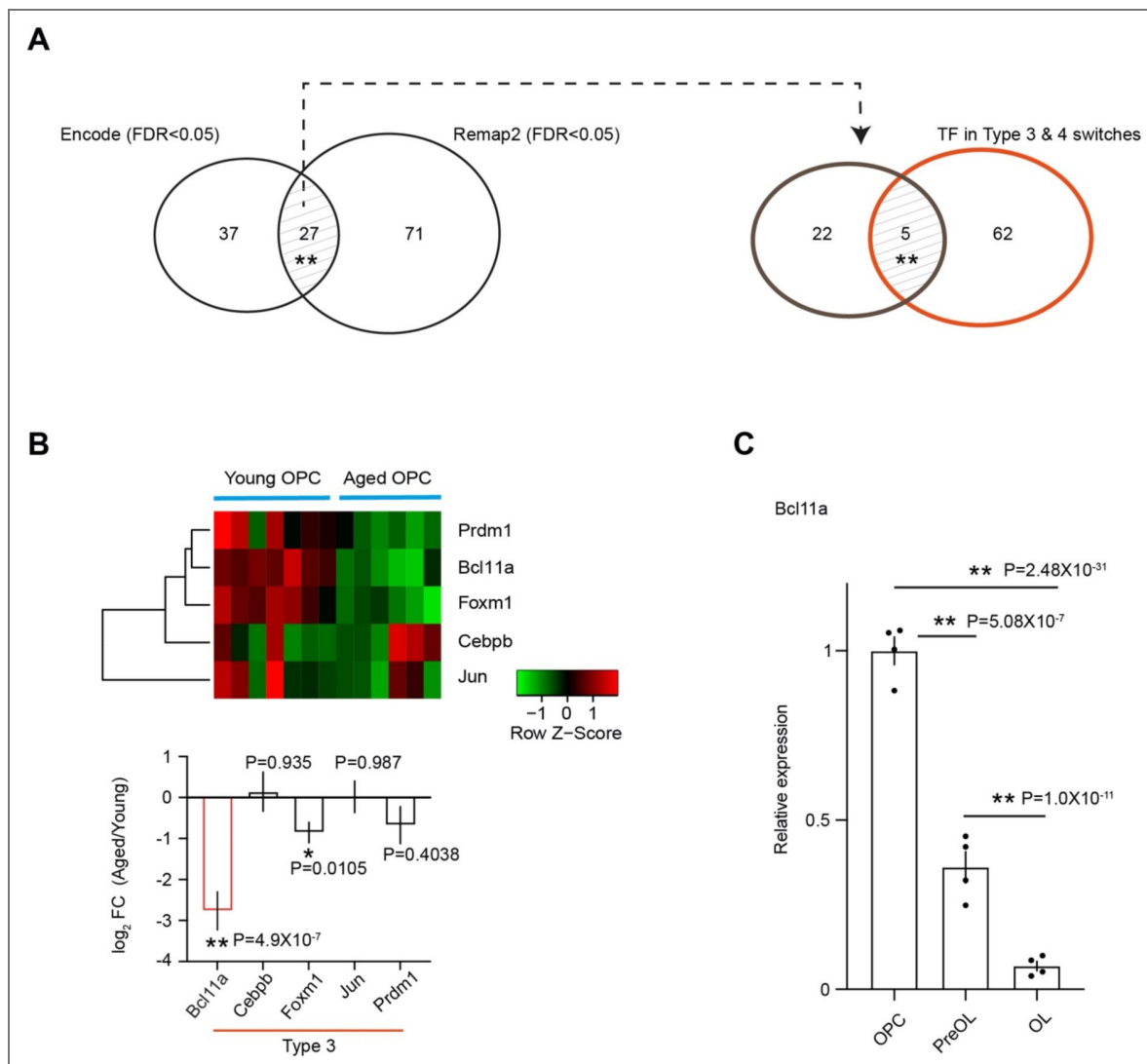


Figure 3. Transcription factor Bcl11a-mediated network potentially affected during ageing.

A) Transcription factor binding site enrichment analysis using Type 3 and Type 4 switch genes as input identified 27 transcription factors (TFs) detected in both ENCODE and ReMap2, which are ChIP-seq-based experimental databases. Among these 27 TFs, we found 5 TFs that are themselves present as Type 3 or Type 4 switches (TF switches extracted from gSWITCH). ** $P < 0.001$, hypergeometric test. **B)** Heat map (top) showing expression (RNA-seq, GEO accession number: GSE303317) of these 5 TFs. Their differential expression in aged OPCs relative to young OPCs plotted as $\log_2 FC$ (bottom). FC: Fold change, $n = 6$ (aged rats), 7 (young rats), mean+SEM, * adjusted p -value < 0.05 , ** adjusted p -value $< 1 \times 10^{-6}$, Benjamini-Hochberg FDR after Wald test. Note: All five TFs are classified as Type 3 switches, and Bcl11a shows the strongest repression in aged OPCs, with the highest statistical significance. **C)** Bcl11a, as a Type 3 TF switch, shows significantly repressed expression as OL lineage progresses. DESeq2-normalised values relative to OPCs plotted. $n = 4$ different rats, mean+SEM, ** adjusted p -value $< 1 \times 10^{-6}$, Benjamini-Hochberg FDR after Wald test. GEO accession number of RNA-seq data: GSE301578.

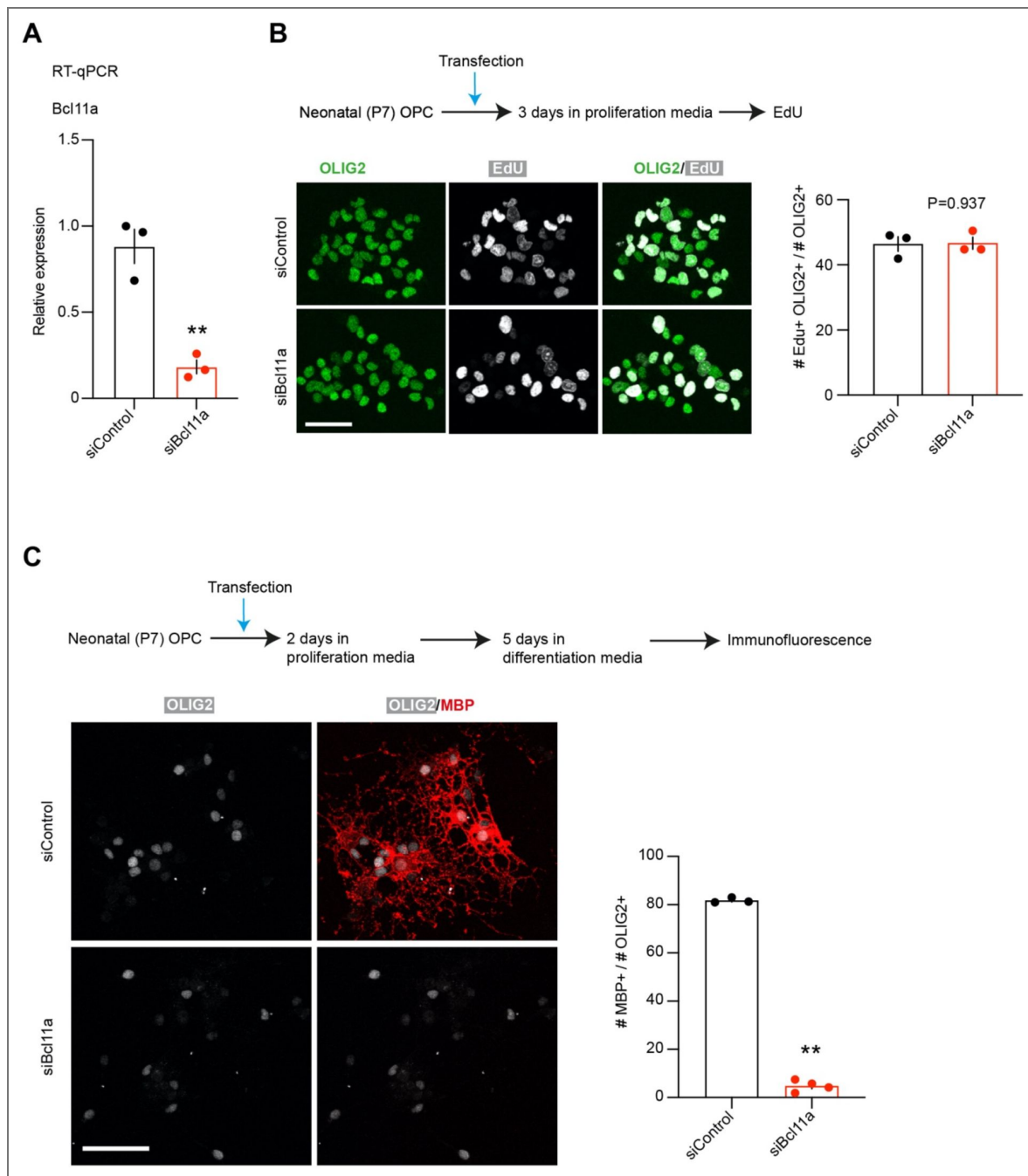


Figure 4. Effect of Bcl11a inhibition on OPCs proliferation and differentiation.

A) Reverse transcription (RT)-qPCR analysis of Bcl11a expression in OPCs transfected with siRNA targeting Bcl11a. siBcl11a: siGENOME Rat Bcl11a; siControl: siGENOME non-targeting control siRNA (Dharmacon™). **B) Effect on proliferation.** Left: Immunofluorescence analysis using antibodies to OLIG2 and EdU detection (Click-iT protocol), followed by EdU proliferation assay. Right: Quantification of EdU+OLIG2+ cells plotted as a percentage of OLIG2+ cells. P: p-value, Student's t test (unpaired and two tailed) **C) Effect on differentiation.** Left: Immunofluorescence analysis using antibody to MBP, OLIG2 upon inhibition of Bcl11a in OPCs. Right: quantification of MBP+ cells were plotted as a percentage of OLIG2+ cells. **(A-C)** n = 3 independent experiments (each time 4 P7 brains were pooled for OPC isolation). mean+SEM. ** p<0.01, Student's t test (unpaired, two tailed) **(B-C)** scale bar: 22 μ m.

Transient overexpression of Bcl11a in aged OPCs improves differentiation

As noted, our RNA-seq analysis revealed reduced Bcl11a expression in aged OPCs (Figure 3B [↗](#)). To investigate whether restoring Bcl11a levels could overcome their impaired differentiation capacity, we expressed Bcl11a in aged OPCs (isolated from 20-24-month-old rats). We first performed *in vitro* transcription to generate Bcl11a mRNA (mBcl11a) and confirmed the fidelity of its protein expression by western blot analysis (Supplementary Figure 2A [↗](#)). mBcl11a was subsequently transfected into aged OPCs; Gfp mRNA (mGfp) served as a control. Transfection efficiency was approximately 77% (Supplementary Figure 2B [↗](#)). Similar to Bcl11a knockdown, mBcl11a expression had no significant effect on aged OPC proliferation (Figure 5A [↗](#)). However, culturing these OPCs in differentiating media led to an approximately 54% increase in OL numbers (Figure 5B [↗](#)), indicating that Bcl11a expression in OPCs is critical for their progression toward the oligodendrocyte lineage.

We next tested the effect of Bcl11a expression in OPCs in aged mice (18-month-old) using a model of focal demyelination in spinal cord white matter. We injected an adeno-associated virus (AAV) carrying a Sox10 driven Bcl11a-P2A-gfp construct into the tail vein of 18-month-old mice (Figure 6A [↗](#)). This design ensures Sox10-driven expression of both Bcl11a and Gfp via a self-cleaving P2A peptide. As a control, we injected AAV carrying a Sox10-driven Gfp construct. One month after AAV injection, we induced spinal cord lesions using lysolecithin. Fourteen days later, the spinal cords were harvested. We found that nearly all GFP immune-positive cells were also OLIG2 positive (98.6% in controls, total GFP+ 4',6-diamidino-2-phenylindole [DAPI]+ cells counted: 193; 98% in Bcl11a, total GFP+ DAPI+ cells counted: 163), indicating strong oligodendrocyte lineage fidelity (Supplementary Figure 2C [↗](#)). Our protocol infected approximately 45% of GFP+OLIG2+ cells (190 GFP+OLIG2+ out of 422 OLIG2+ cells counted in control; 160 GFP+ OLIG2+ out of 357 OLIG2+ cells counted in Bcl11a) (Supplementary Figure 2C [↗](#)). We performed RNA *in situ* hybridization using a probe specific for proteolipid protein 1 (Plp1), a marker of late oligodendrocyte differentiation. We observed no difference in total OLIG2+ cells in the lesion area but a significant increased number of OLIG2+ cells that were Plp1+ in Bcl11a samples (Supplementary Figure 3 [↗](#); Figure 6B [↗](#)), suggesting Bcl11a restores the differentiation capacity of aged OPCs recruited to the injury area.

Bcl11a has undergone purifying selection during vertebrate evolution

Given that Bcl11a functions as an important regulatory transcription factor switch during oligodendrocyte (OL) lineage progression and that its expression level is associated with functional vulnerability in ageing, we used computational molecular evolution approaches to investigate whether selective constraints during evolution have acted on its codon sequence. We aimed to determine whether such constraints have preserved the amino acid sequence by eliminating deleterious mutations through purifying selection. Bcl11a is absent in invertebrate genomes (UCSC Genome Browser search), but orthologs are present across all examined vertebrate species. The ratio of nonsynonymous (dN) to synonymous (dS) substitution rates — known as the dN/dS ratio — is a widely used metric in molecular evolution to evaluate the selective pressures on protein-coding genes (Yang Z, 2007 [↗](#), Anisimova M and Kosiol C, 2009 [↗](#)). As shown in Table 1 [↗](#), our analysis using multiple evolutionary models consistently yielded dN/dS values below 1, providing strong evidence of pervasive purifying selection acting on Bcl11a across the vertebrate lineage.

Discussion

Dynamically regulated genes during transitions from one cellular stage to another offer important information about the regulatory mechanisms governing differentiation (Spitz and Furlong, 2012 [↗](#); Pope and Medzhitov, 2018 [↗](#)). Our gSWITCH web application is a powerful tool for identifying specific dynamic patterns based on progressive changes in gene expression during cellular state transitions or any variable-dependent changes. This can be particularly valuable for

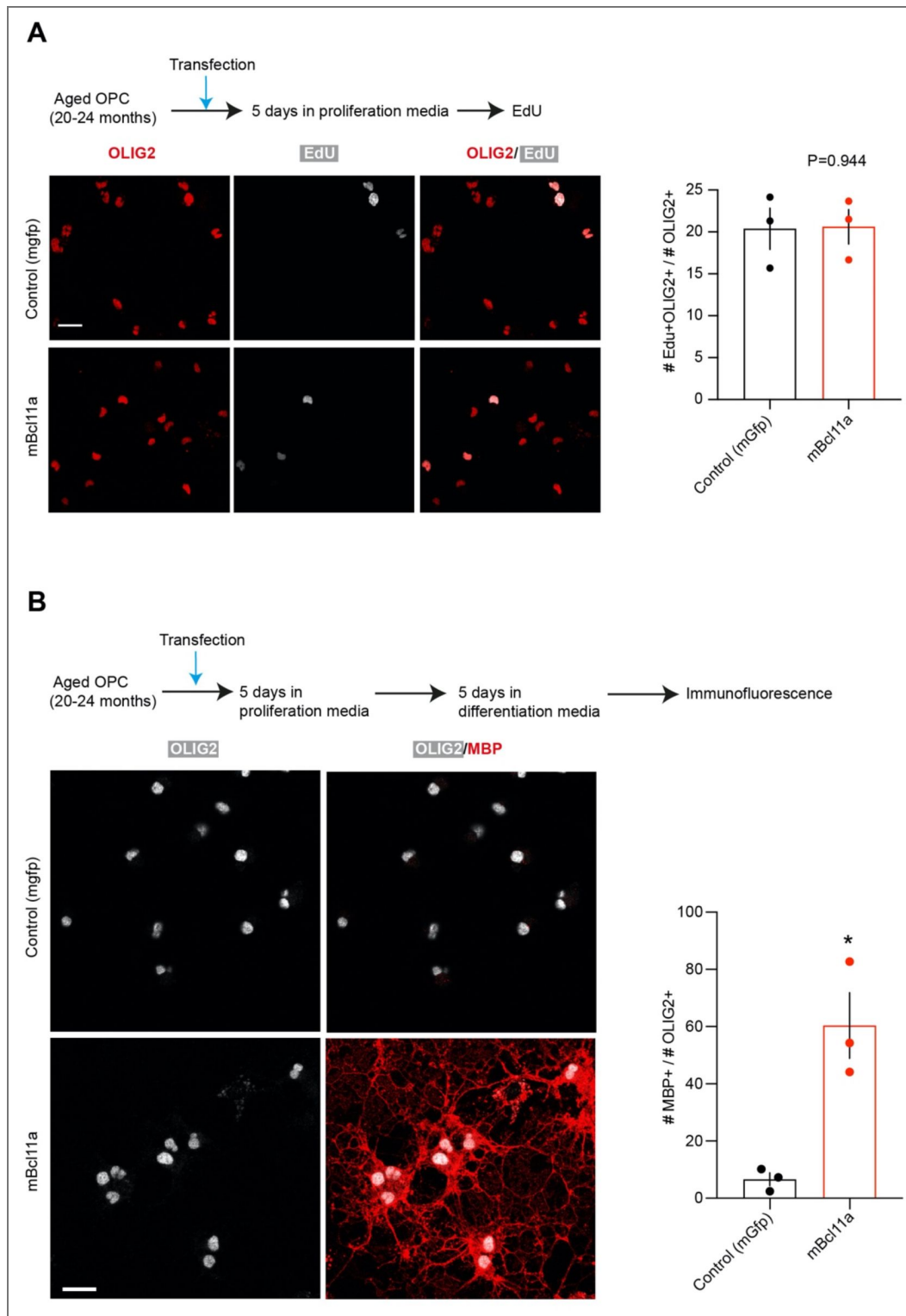


Figure 5. Effect of Bcl11a transient over-expression on proliferation and differentiation of aged OPCs.

A) Effect on proliferation. Left: Immunofluorescence analysis using antibodies to OLIG2 and EdU detection (Click-iT protocol), followed by EdU proliferation assay. Right: Quantification of Edu+OLIG2+ cells plotted as a percentage of OLIG2+ cells. P: p-value, Student's t test (unpaired, two tailed). **B) Effect on differentiation.** Left: Immunofluorescence analysis using antibodies to MBP and OLIG2 after transfection of Bcl11a mRNA (mBcl11a) or Gfp mRNA (mgfp; control) in aged OPCs. Right: Quantification of MBP+ cells plotted as a percentage of OLIG2+ cells. * P<0.05, Student's t test (unpaired, two tailed). **(A-B)** n = 3 independent experiments (each using OPCs isolated from pooled brains of 2 aged rats). mean ± SEM. scale bar: 22 µm.

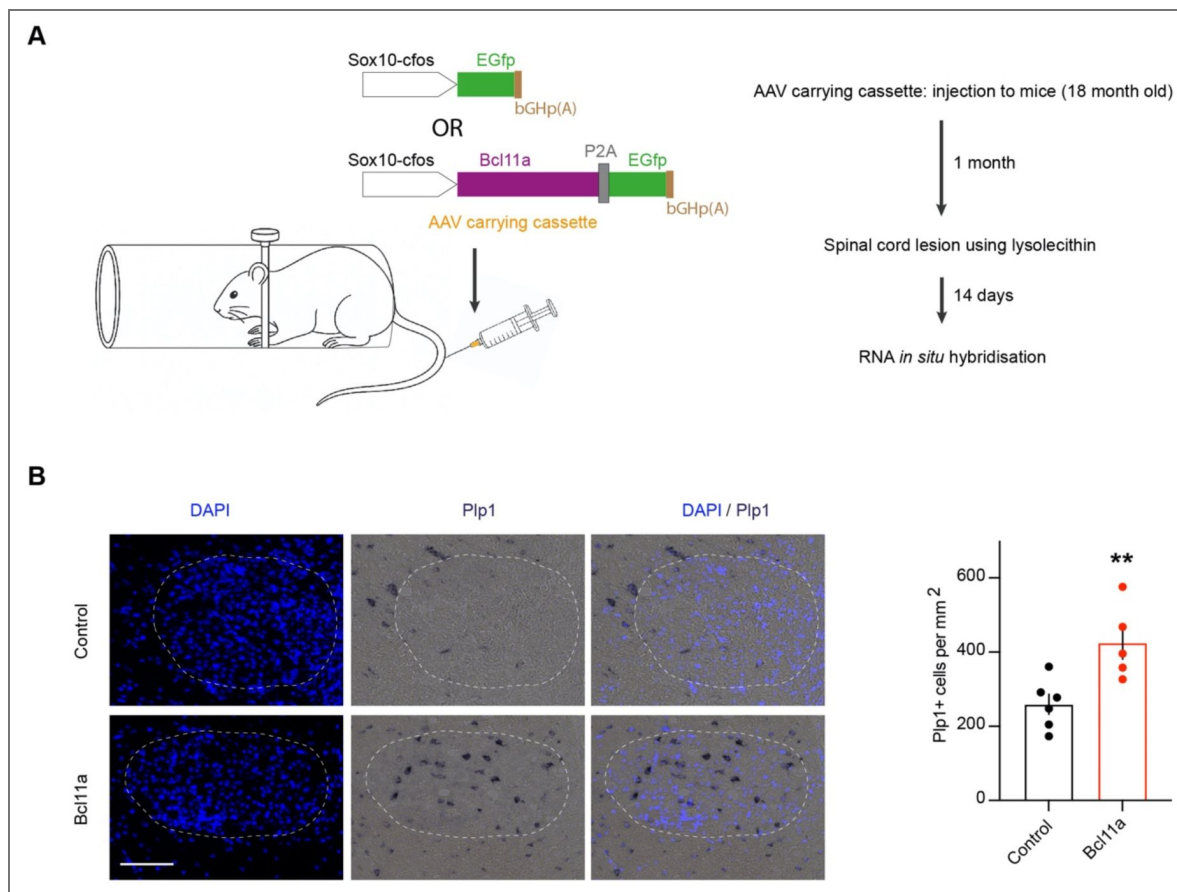


Figure 6. OL lineage specific Bcl11a expression in aged mice.

A) Left: SOX10-driven construct carrying either Gfp alone (control) or Bcl11a linked to Gfp via a self-cleaving peptide element (P2A). This construct ensures SOX10-driven expression of both Bcl11a and Gfp. P2A: Porcine teschovirus-1 2A peptide, bGH p(A): bovine growth hormone poly(A) signal. The AAV containing this construct was delivered to 18-month-old mice via tail-vein injection. Right: schematic showing the experimental timeline. **B**) RNA *in situ* analysis of Plp1. In the lesioned area, the number of Plp1⁺ cells per mm² was counted and plotted. n=5-6 mice, mean±SEM, ** P<0.01, Student's t test (unpaired, two tailed). scale bar: 100 μm. The white dotted line demarcates the lesion area.

Model	Description	Parameter estimates and test statistics	Interpretation
M0	M0 (one ratio): assumes that all sites have the same dN/dS.	dN/dS= 0.064 (SE: 0.0023) logL= -21196.13	M0 suggests pervasive purifying selection
M3 compared with M0	M3 (discrete): permits 3 different types of sites with different dN/dS values. Comparison of models M0 and M3 is a test of whether there is variation of selective pressures over sites, including the possibility of sites with dN/dS > 1.	Compared with M0: dlogL = -20788.43 - (-21196.13)= 407.7 χ^2 test (4 df) indicates significant preference for M3 (P < 0.00001)	LRT indicates M3 significantly preferred to M0, i.e. strong evidence of selective pressures varying over sites. All sites under purifying selection
M2a compared with M1a	M1a (neutral): assumes 2 types of sites, one with dN/dS < 1 and one with dN/dS = 1. M2a (selection): in addition to the categories in M1a, includes a third class of sites with dN/dS > 1. Comparison of these models is a test of whether there are any sites with dN/dS > 1.	dlogL = -20983.51- (-20983.62) = 0.11 χ^2 test (2 df) indicates significant preference for M1a (P = 0.90)	Absence of positive selection.

Note: dN: number of non-synonymous changes per non-synonymous site, dS: number of synonymous changes per synonymous site, SE: Standard error, logL: log-likelihood value, dlogL: log-likelihood difference, LRT: likelihood ratio test (2dlogL compared with appropriate χ^2 distribution).

Table 1. Test for evolutionary selection on Bcl11a.

discovering novel mechanisms in defined biological contexts. Switch genes signify the neurogenic commitment of cortical progenitors and have enabled the identification of new regulators of brain corticogenesis (Aprea et al., 2013). However, no biologist-friendly tool has been available to systematically detect such genes. gSWITCH is a biologically inspired method, and here we experimentally validate its prediction.

Using gSWITCH and following a data driven approach, we identified Bcl11a as a critical switch gene in the transition from OPCs to pre-OLs. To test this prediction, we performed loss-of-function (LOF) experiments using siRNA and found that inhibition of Bcl11a impaired OPC differentiation into oligodendrocytes. Because Bcl11a expression is markedly down-regulated in aged OPCs, we next carried out gain-of-function (GOF) experiments both *in vitro* and *in vivo*. Restoring Bcl11a expression in aged OPCs significantly enhanced their differentiation. Together, the LOF and GOF experiments unambiguously validate the key role of Bcl11a in oligodendrocyte lineage progression and support the gSWITCH prediction.

Ageing is undeniably complex, with multiple cellular, metabolic, and extracellular changes, inflammation occurring in parallel. Discovering new regulators remains essential for understanding how these layers collectively shape age-related decline. Interventions such as fasting (Neumann et al., 2019), modulation of the extracellular matrix (Segel et al., 2019), or over-expression of transcriptional regulators like cMyc (Neumann et al., 2021) have shown that no single mechanism can fully explain the functional deterioration of ageing oligodendrocyte progenitors. Instead, each factor highlights a distinct point at which the ageing process exerts pressure, gradually revealing the distributed architecture through which ageing unfolds. Within this landscape, our identification of Bcl11a—emerging from an analysis of oligodendrocyte lineage progression—defines another biologically meaningful regulatory axis involved in these age-associated changes, expanding the mechanistic resolution with which ageing can be understood. Only by charting these individual components can we explain how diverse age-related diseases emerge and begin to design personalised strategies capable of addressing the true complexity of ageing in a mechanistically grounded way.

BCL11A is a zinc-finger transcription factor primarily known as a transcriptional repressor, though it can also function as an activator (Horton et al., 2025; Bauer and Orkin 2015; Lee et al., 2017). Our computational molecular-evolution analyses indicate that Bcl11a is a vertebrate-specific transcription factor that has undergone pervasive purifying selection. In this context, Bcl11a is perhaps evolutionarily linked to major cellular programs, and its dysfunction may contribute to disease. In haemoglobin regulation, Bcl11a orchestrates the fetal-to-adult haemoglobin switch by binding to and repressing the γ -globin genes, thereby enabling activation of adult β -globin—a critical transition that supports the oxygen-transport requirements of postnatal life (Liu et al., 2018). In oligodendrocytes, we find that Bcl11a is essential for lineage progression. Bcl11a expression in young OPCs is important for maintaining their functional resilience, ensuring they retain a strong capacity to differentiate efficiently.

Notably, Bcl11a deficiency or dysfunction has been implicated in autism spectrum disorder and intellectual disability based on human genetic studies (De Rubeis et al., 2014; Dias et al., 2016; Cai et al., 2017). In the hematopoietic system, Bcl11a deficiency in hematopoietic stem cells induces an ageing-like state (Luc et al., 2016). Consistent with this, we observed that aged OPCs, despite being impaired in their differentiation capacity, were able to regain functional resilience when Bcl11a expression was restored. In this sense, Bcl11a reinstates functional robustness in aged OPCs. Future studies will elucidate the detailed gene-regulatory mechanisms through which Bcl11a functions in young OPCs, clarify how it enhances functional competence in aged OPCs, and further define its potential as a therapeutic target.

Materials and Methods

gSWITCH

[This includes section g.1 to g.5]

g.1 Usage

It is designed to handle gene-expression data from lineage progression or time series or any variable dependent experiments. Experiment type: bulk RNA-seq, single cell RNA-seq experiments flattened to pseudo bulk. In the series data, the number of series or time points or group ≥ 3 and the replicates per time point or group ≥ 2 .

g.2 Access: freely available at <https://altoslabs.shinyapps.io/gSWITCH/> 

g.3 Implementation

gSWITCH is a web application and operating system (OS) independent. gSWITCH is implemented in 'R' by using *reshape2*, *dplyr*, *DESeq2*, *ggplot2*, *tximport*, *MonoInc* and *plotly* (R Core Team, 2019 ; Wickham, 2007 ; Wickham, 2016 ; Wickham, 2017 , Love et al., 2014 ) and employing RStudio's *shiny* and *shinyrcssloaders* packages for web application.

g.4 Logic

gSWITCH is based on the following: 1) GLM (Generalised linear model) fit and hypothesis testing, 2) specific contrasts analysis, 3) absolute maximum/minimum (in a closed interval) and monotonicity to categorize switch types.

We refer the Supplementary Figure 1 . We employ DESeq2 for GLM fit, calculation of LTR (likelihood ratio test) p-value (adjusted by FDR) and select genes for which adjusted p value (Adj p) < user-defined-value (u.d.) [0.05 is set as default]. Let say these genes are our set (i). We then calculate contrast between the first and the last cell state or time points and the genes for which Adj p < user defined (or 0.05 as default) and the magnitude of fold change $|FC| > \text{user-defined-value}$ (or 1.4 is set as default) was considered as set (ii). Intersection of (i) and (ii) was taken to identify four different switch types which are based on absolute max/min and monotonicity. A relative expression value (g) was calculated by determining expression level of each cell state relative to the mid-point [for n=odd number: mid-point= $\{(n/2)+0.5\}$ th cell state; for n=even number: mid-point= $(n/2)$ th cell state].

1. Type 1: properties: (a) absolute maxima at the end point (b) 'g' is monotonically increasing i.e. within $[g_1, g_n]$, $g_{j-1} < g_j$ ($j=1,2,\dots,n$) n is a positive integer ≥ 3 (c) contrast analysis: X_n vs X_{n-1} : Adj p < u.d. (or 0.05), $|FC| > 1.4$ (or u.d.).
2. Type 2: properties: (a) absolute maxima at the mid-point (b) 'g' is monotonically increasing from start to mid-point and then monotonically decreasing from the mid-point to end point i.e. within $[g_1, g_j]$, $g_j \geq g_{j-1}$ AND $[g_j, g_n]$, $g_j \geq g_{j+1}$ (c) contrast analyses: X_j vs X_{j+1} AND X_{j-1} vs X_j : Adj p < 0.05 (or u. d.), $|FC| > 1.4$ (or u.d.).
3. Type 3: properties: (a) absolute maxima at the start point (b) 'g' is monotonically decreasing as it progresses to end point i.e. within $[g_1, g_n]$, $g_{j-1} \geq g_j$ (c) contrast analysis: X_1 vs X_2 : Adj p < 0.05 (or u.d.), $|FC| > 1.4$ (or u.d.).
4. Type 4: properties: (a) absolute minima at the mid-point (b) 'g' is monotonically decreasing from start to the mid-point and then monotonically increasing from the mid-point to the end point i.e. within $[g_1, g_j]$, $g_{j-1} \geq g_j$ AND $[g_j, g_n]$, $g_j \leq g_{j+1}$. (c) contrast analyses: X_j vs X_{j+1} AND X_{j-1} vs X_j : Adj p < 0.05 (or user defined), $|FC| > 1.4$ (or u.d.).

Identifying transcription factors (TFs)

Comprehensive list of TFs were retrieved from TFcheckpoint database (Chawla K et al., 2013 ) , an experimental evidence-based database. L $\cap$ S is the TFs in the switch genes. L: list from TFcheckpoint, S: switch genes.

g.5 Parameters used in this paper

Adj p < 0.05, FC = 1.5

RNAseq analysis

RNA was isolated using RNA MiniPrep (Zymo Research, D7001) following manufacturer protocol. Dnase treatment on column (as per manufacturer protocol) was included in this isolation protocol. Paired end directional library was prepared using NuGen kit: Ovation RNA-seq Systems 1-16 for

model organisms (0349-32).

Oligodendrocyte lineage dataset: The accession number of this dataset is **GSE301578**. Data were pre-processed through sortMeRNA, trimmomatics. Rat transcriptome (Rnor_6.0.cdna+Rnor_6.0.ncrna) index was built and quantification was performed using Salmon (Quasi mode). Data was normalised using DESeq2. This tximport of salmon quantification were run on gSWITCH (accessible at <https://altoslabs.shinyapps.io/gSWITCH/>) for switch genes identification and visualisation.

Analysis of aged and young Rat OPC data: The accession number of this dataset is **GSE303317**. Data were pre-processed using sortMeRNA and Trimmomatic. The rat transcriptome index (Rnor_6.0.cdna + Rnor_6.0.ncrna) was built, and quantification was performed using Salmon in quasi-mapping mode. Raw Salmon counts were imported using the tximport package. Differential expression analysis was conducted with DESeq2, incorporating batch as a covariate in the design formula: design = ~ batch + group. P-values were adjusted for multiple testing using the false discovery rate (FDR) method. Genes with an adjusted p-value less than 0.05 and a fold change greater than or equal to 1.5 were considered differentially expressed between the two groups. For data visualization, counts were normalized using a variance stabilizing transformation (VST) (varianceStabilizingTransformation function from DESeq2), and batch effects were removed using the removeBatchEffect function from the limma package (Ritchie et al., 2015).

TF-binding site enrichment analysis

Using type 3 and type 4 switches as input, we employed the ChIP-X Enrichment Analysis (ChEA3) tool [<https://maayanlab.cloud/chea3/>] (Keenan et al., 2019) and utilised two experimentally curated ChIP-seq databases—ENCODE (Encyclopedia of DNA Elements) (ENCODE Project Consortium, 2012) and ReMap2 (Regulatory Map database) (Hammal et al., 2022)—to identify transcription factors (TFs) with significantly enriched binding sites (FDR < 0.05). We then focused on transcription factors consistently identified by both ENCODE and ReMap2.

dN/dS analysis

Twenty-three vertebrate species (including lamprey), spanning a broad range of vertebrate phyla, were used in this study. The longest (XL) coding sequences (CDS) and the corresponding protein sequences of Bcl11a were obtained from Ensembl (Dyer et al., 2025). For lamprey, the sequences were retrieved from GenBank (Sayers et al., 2021), as they were not available in Ensembl. Protein sequences were aligned using MAFFT (Katoh and Standley, 2013), and the resulting alignment together with the corresponding coding sequences was used as input to the PAL2NAL program (Suyama et al., 2006) to generate codon-based nucleotide alignments. The species tree was generated from CDS using the NGPhylogeny server (Lemoine et al., 2019), the nucleotide sequence alignment, BMGE (Criscuolo and Gribaldo, 2010) for alignment curation, and PhyML (Guindon et al., 2010) for tree inference. dN/dS ratios and likelihood ratio tests of model comparisons were performed using Codeml in the PAML package v4.8a (Yang, 2007). Codon frequency was set to F3X4, the universal genetic code was specified in the icode variable, and the program was run with the following selection model parameters: Nssites 0 and 3 (selection models M0 and M3), Nssites 1 and 2 (models M1a and M2a), and Nssites 7 and 8 (models M7 and M8). Significance of log-likelihood differences for model comparisons was tested using χ^2 statistics with appropriate degrees of freedom, as described in the PAML manual (<https://github.com/abacus-gene/paml/blob/master/doc/pamlDOC.pdf>).

Further detail on the M7 and M8 comparison: these models differ only by M8 allowing an additional class of sites with dN/dS > 1. The likelihood ratio test (LRT) comparing these indicates a significant preference for M8 (P = 0.006) but identifies only a single site with posterior mean dN/dS > 1 and confirms all other sites to be evolving under purifying selection. (Codon 156, i.e. nucleotide positions 466–468 in our curated alignment, is inferred to have dN/dS = 2.93 ± 2.81 and posterior probability 0.94 of being under positive selection. However, inspection of the alignment reveals that this position is represented only in macaque, rabbit, blue tit, and frog, making evidence for positive selection across species unconvincing even at this one site.)

Animals

Sprague–Dawley rats (C. River, Margate, UK) and C57Bl/6 mice (18-month-old) were housed in individually ventilated cages at $22\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$ and $60\% \pm 5\%$ humidity under a 12-h light–dark cycle. Food and water were provided ad libitum in a standard rodent facility at the University of Cambridge, UK or Altos Labs-Cambridge Institute of Science, UK. All procedures were carried out in accordance with the Animals (Scientific Procedures) Act 1986 (Amendment Regulations 2012) and were approved by the Animal Welfare and Ethical Review Body (AWERB) of the University of Cambridge and Agenda Resource Management.

Cellular isolation

Rats were sacrificed using an overdose of pentobarbital and the whole brain was dissected immediately in ice cold Hibernate A low fluorescence (HALF) medium (PMID: 31585093). Subsequently cellular isolation was performed as described. Per 10 million cell suspension, we used $2.5\text{ }\mu\text{g}$ of primary antibody. Oligodendrocytes (OL) were isolated first using goat-anti-mouse-MOG-Biotinylated (RD systems, BAF2439); followed by OPC were isolated from the MOG negative population using mouse-anti-rat-A2B5-IgM antibody (Millipore; MAB312). Finally, pre oligodendrocytes (PreOL) were isolated from MOG and A2B5 negative population using mouse-anti-O4-IgM (R&D systems, MAB1326). The respective secondary antibodies (20 μl) were: mouse-anti-biotin magnetic micro beads (Miltenyi, 130-105-637), rat-anti-mouse-IgM antibody magnetic beads (Miltenyi, 130-047-302), rat-anti-mouse-IgM magnetic beads (Miltenyi; cat#130-047-302).

Cell culture and differentiation

Culture of aged OPCs (isolated from 20–24-month-old rats) were performed as described (PMID: 31585093). Additionally, DMEM-F12 (Sigma), GlutaMAXTM (GibcoTM) and B27 supplement (GibcoTM) has been used in the media for aged OPC culture. Proliferation medium contains PDGF and bFGF at 30 ng/ml each (PeproTech). For differentiation as described (Ghosh et al., 2024 [DOI](#)) OPC medium was supplemented with 40 ng ml^{-1} 3,3',5-Triiodo-L-thyronine (T3) (Sigma; T6397). Medium was replaced on alternate days for 5 days.

Neonatal OPCs (isolated from P7 rats) were cultured as described (Ghosh T et al., 2024 [DOI](#)).

Cloning of Bcl11a

The original Human XL BCL11A clone was a gift from Prof. Walid Khaled lab (Cambridge University, UK). BCL11A was subcloned to a TOPO vector (pCRTM 2.1-TOPOTM TA using TOPOTM TA CloningTM Kit following manufacturer instruction) and sequence was verified. For *In vitro* transcription, the following primers were used for PCR amplification of BCL11A:

FP: 5' TAATACGACTCACTATAGgccaccATGTCTCGCCGCAAGCAA 3'

(italics: T7 promoter sequence, small case: kozac sequence)

RP: 5' CTATTCAGTTTTTATATCATTATTCAACACTCGATCA 3'

Purified PCR product (from above) was used as a template and transcribed *in vitro* using HiScribeTM T7 ARCA mRNA Kit (New England Biolabs) with tailing and modified nucleotides (Tebu Bioscience) following manufacturer protocol (E2060S). mRNA was purified using DirectzolTM RNA Miniprep kit (Zymo Research). CleanCap^R EGFP mRNA (L-7601) was procured from Tebu Bioscience.

Western blot analysis

MCF7 cells were used to verify protein-level expression of BCL11A following transfection with *in vitro*–transcribed Bcl11a mRNA (see above). MCF7 cells were selected because they exhibit poor endogenous BCL11A expression (Waterhouse M et al., 2025 [DOI](#)). For mRNA transfection method (see below). After 36 hours of transfection, cells were lysed in the wells using RIPA buffer containing protease inhibitors. The lysates were incubated at $4\text{ }^{\circ}\text{C}$ for 30 min and centrifuged at maximum speed at $4\text{ }^{\circ}\text{C}$ for 30 min. The resulting supernatant was used for protein separation on a 7.5% SDS–

PAGE gel and subsequently transferred to a PVDF membrane for Western blotting. Antibodies used were, anti-BCL11A (ab191401, Abcam, 1:3000) (Waterhouse et al., 2025 [↗](#)) and an HRP-conjugated donkey anti-rabbit IgG secondary (NA934, Amersham/GE Healthcare) (1:10000); anti-beta Actin antibody (ab6276) (1:3000) and HRP-conjugated sheep anti-mouse IgG secondary (NA931, Amersham/GE Healthcare) (1:10000). Immunoreactive bands were detected using ECL reagent (Amersham) and imaged an Odyssey Imaging system.

Transfection

siRNA transfection in neonatal OPCs: We used LipofectamineTM RNAiMAX transfection reagent (ThermoFisher scientific) and follow the protocol as described (Ghosh et al., 2024 [↗](#)). siRNAs were procured from DharmaconTM. For Bcl11a we used siGENOME Rat Bcl11a (305589)-SMARTpool (Cat no. M-085668-01-0010); for control we used siGENOME non-Targeting siRNA Pool #2 (Cat no. D-001206-14-05)

mRNA transfection in aged OPCs: 60,000 cells per well of a 24 well plate was seeded. Cells were kept in proliferation media [PDGF and bFGF, 30ng/ml each (Peprotech)] for 5 days. mRNA (approx. 500ng) was transfected using LipofectamineTM MessengerMaxTM transfection reagent (ThermoFisher scientific) following manufacturer protocol.

Immunofluorescence

Cells were fixed in 4% paraformaldehyde (PFA) for 10 min at room temperature, washed in PBS, and processed for immunofluorescence staining as described previously (Neumann et al., 2019 [↗](#)). For in vivo experiments, animals were perfused with 4% PFA in PBS; spinal cords were post-fixed overnight in 4% PFA, cryoprotected in 20% sucrose, and embedded in OCT medium (Tissue-Tek). Immunofluorescence staining was performed on 12 µm cryostat sections following previously described method (Rawji et al., 2018 [↗](#)).

The antibodies used in this study, described previously (PMID: 38364788), were: rat anti-MBP (MCA409S, Serotec) [1:500], rabbit anti-OLIG2 (AB9610, Millipore) [1:1000 or 1:100], chicken anti-GFP (ab13970, Abcam) [1:1000], donkey CyTM3 AffiniPure anti-Rat (712-165-150, Jackson Immuno Research) [1:1000], donkey Alexa Fluor 647 anti-Rabbit (A31573, Invitrogen) [1:1000 or 1:500], donkey Alexa Fluor 488 Anti-Chicken (703-546-155, Jackson Immuno Research) [1:1000].

Images were acquired using SP8 (Leica TCS) confocal microscope. All acquisition settings were kept constant throughout the experiment. Images were processed in ImageJ version 2.1.0 (Schindelin et al., 2012 [↗](#)). Cell counting was performed using the Cell Counter tool in ImageJ. Immunofluorescence experiments were performed and analysed blinded to treatment.

EdU proliferation assay

For EdU incorporation, cells were incubated with 10 µM EdU for 5 hours and subsequently processed following the manufacturer's instructions (Click-iT Plus EdU Alexa Fluor 647 Imaging Kit, Thermo Fisher, C10640). In vitro tissue-culture cells were fixed in 4% paraformaldehyde (Thermo Fisher, 10131580) for 10 minutes at room temperature. Cells were washed once with PBS and blocked in PBS containing 0.1% Triton X-100 and 5% donkey serum (Sigma, D9663) for 30 minutes at room temperature. EdU detection was performed according to the kit protocol, followed by immunostaining. Antibodies used were: Anti-OLIG2: rabbit anti-OLIG2 (AB9610, Millipore) [1:1000], Goat Alexa Fluor 488 anti-rabbit IgG (A11008, Invitrogen) (1:1000) [siRNA experiments in Figure 4 [↗](#)], Goat Alexa Fluor 594 anti-rabbit IgG (A11012, Invitrogen) (1:1000) [mRNA transfection experiment in Figure 5 [↗](#)].

RT-qPCR

Total RNA was isolated using the mirVana kit (Ambion). RNA samples were treated with DNase I (Thermo Fisher Scientific) prior to reverse transcription. cDNA was generated using random hexamers (Invitrogen) and SuperScript IV Reverse Transcriptase (Invitrogen) according to the manufacturer's protocol. Real-time PCR was performed using SYBR Green Master Mix (Applied

Biosystems) on a QuantStudio 7 Flex system (Applied Biosystems). The PCR program was: 50 °C for 2 min, 95 °C for 10 min, followed by 40 cycles of 95 °C for 15 s and 60 °C for 1 min. A melt-curve (dissociation) analysis was performed at the end.

Relative quantification of qPCR data was performed using the Pfaffl method (Pfaffl MW, 2001 [↗](#)). PCR efficiencies for the target gene (Bcl11a) and the endogenous control (Actb) were determined from the slopes of their respective standard curves.

Rat specific primers:

Bcl11a:

F: GTGACACCAGAAGATGACGA

R: ATTAGGGCTCCGTGTGCAG

[Amplicon: 131 bp]

Actb:

F: CACCATGTACCCAGGCATTG

R: CACACAGAGTACTTGCGCTC

[Amplicon: 111 bp]

AAV delivery

The SOX10 multiple-species conserved enhancer element 5, conjugated to the *cfos* basal promoter (Addgene #115783; Pol et al., 2013 [↗](#)), was used to drive oligodendrocyte-lineage-specific expression of Bcl11a and EGfp (Figure 6A [↗](#)). For the control construct, the same regulatory sequence drove expression of EGfp alone. The entire SOX10-driven expression cassette (Figure 6A [↗](#)) was cloned between the inverted terminal repeats (ITRs) of an adeno-associated virus 2 (AAV2) vector. Cloning, large-scale AAV-PHP.eB packaging, and purification were performed by GenScript (UK).

AAV carrying the construct (Figure 6A [↗](#)) was administered to 18-month-old mice via tail vein injection at a dose of 5×10^{11} genome copies (GC) in a total volume of 200 μ L.

Focal demyelination of the spinal cord

A mouse spinal cord demyelination model was generated by injecting 1% lysophosphatidylcholine (lysolecithin, LPC; Sigma-Aldrich, Gillingham, UK) prepared in PBS into the ventral white matter of the spinal cord at the Th13/L1 level following surgical exposure under isoflurane anaesthesia. 18-month-old C57Bl/6 mice were used in this study; the animals do not manifest abnormalities in this model. Mice were sacrificed by perfusion fixation at 14 days post lesioning (dpl) for histological analysis.

Plp1 *in situ* hybridisations (ISH)

We followed standard procedures used in the Richardson laboratory (UCL) (<https://www.homepages.ucl.ac.uk/~uchzwdr/In%20Situ%20Protocol.pdf> [↗](#)). Plp1 mRNA was detected using a digoxigenin (DIG)-labelled riboprobe. Primers (FP: GGGGATGCCTGAGAAGGT; RP: TGTGATGCTTTCTGCCCA) targeting the Allen Brain Atlas Plp1 sequence (ABA experiment 75496148) were used to amplify the region, which was subsequently TA-cloned into the pCRII vector using the Thermo Fisher K206001 kit. Insert orientation was confirmed by Sanger sequencing. Following linearisation with XbaI, an antisense DIG-labelled riboprobe (970 bp) was synthesised by *in vitro* transcription using T7 RNA polymerase.

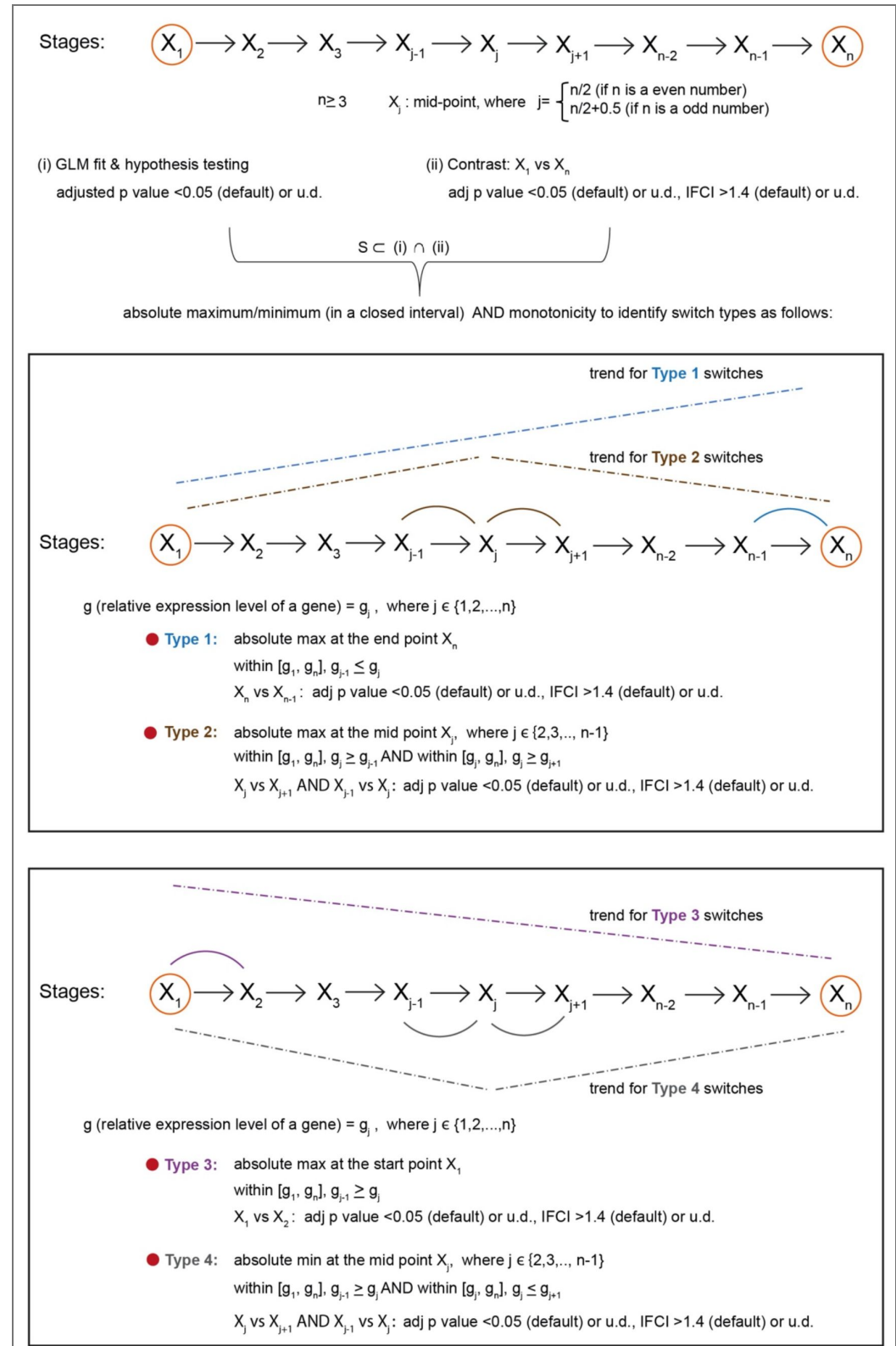
For hybridisation, the DIG-labelled probe was diluted 1:1000 in hybridisation buffer [50% de-ionised formamide, 1 $\times$ salts (0.2 M NaCl, 5 mM EDTA, 10 mM Tris-HCl pH 7.5, 5 mM NaH₂PO₄, 5 mM Na₂HPO₄), 0.1 mg/ml yeast tRNA, 10% dextran sulphate, and 1 $\times$ Denhardt's solution], denatured at 70–75 °C for 10 min, and applied to tissue sections. Slides were incubated overnight at 65 °C in a humidified chamber containing 50% formamide/1X SSC. The next day, sections were washed in three pre-warmed solutions of 50% formamide/1X SSC/0.1% Tween-20 at 65 °C, followed by two

room-temperature washes in 1X MABT (100 mM maleic acid, 150 mM NaCl, 0.1% Tween-20, pH 7.5). Sections were blocked for 1 h in 20% sheep serum and 5X blocking reagent (2% Roche blocking reagent + 10% heat-inactivated sheep serum) in 1X MABT, then incubated overnight at 4 °C with anti-DIG-AP antibody (1:1000) prepared in the same blocking solution. After three washes in MABT, colour development was performed using NBT/BCIP in a freshly prepared Tris/MgCl₂/NaCl developing buffer with 5% polyvinyl alcohol (PVA) at pH 9.8, incubated at 37 °C in the dark until optimal signal intensity was achieved. Slides were rinsed in PBS, counterstained with DAPI, and mounted in Fluoromount™ (Sigma) and imaged under Thermo Fisher EVOS M5000 Cell Imaging System. ISH experiments were performed and analysed blinded to treatment.

Statistical analysis

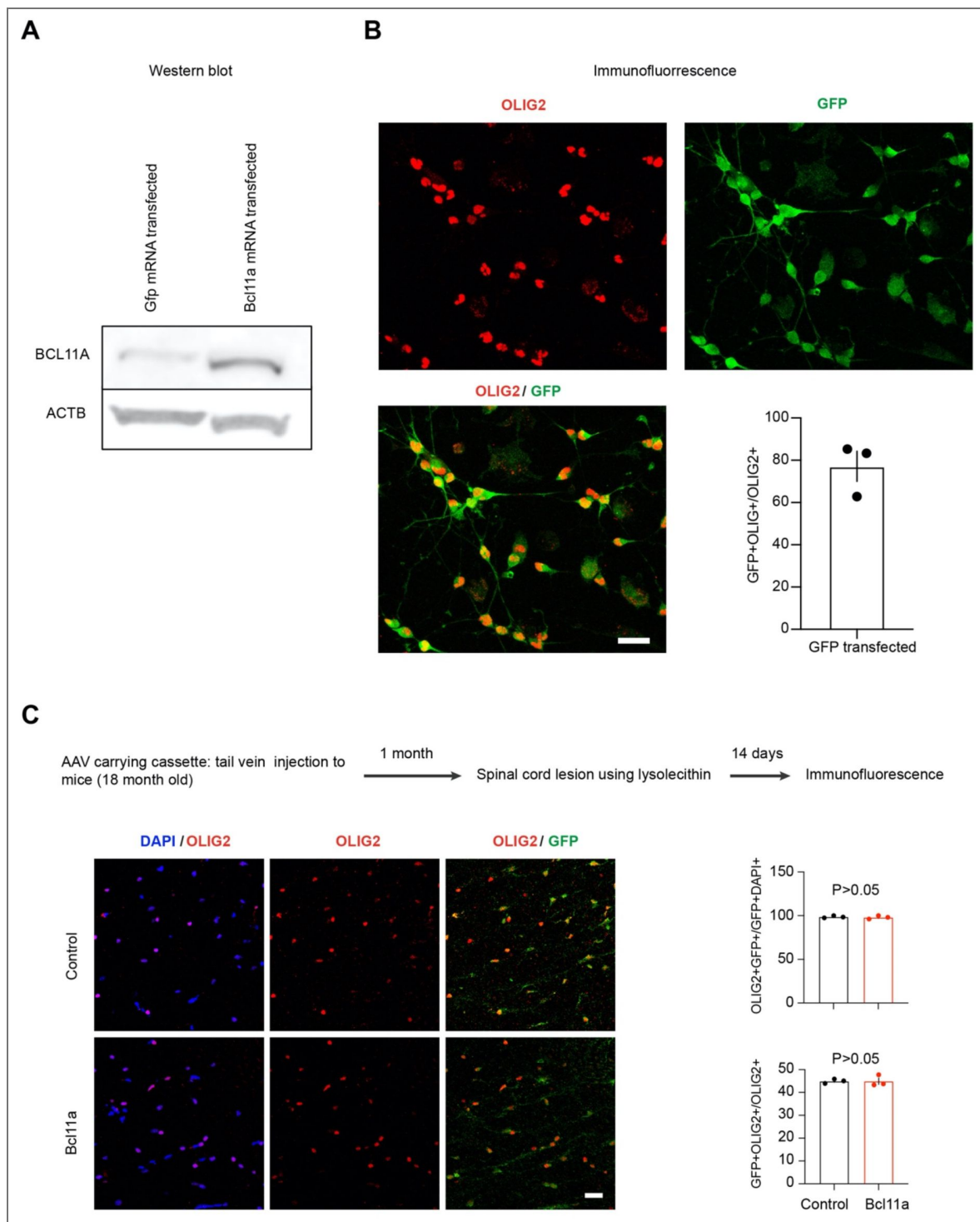
R (version 4.0.2) or GraphPad Prism (version 8.4.3) was used for statistical analyses. Normality was evaluated using the Shapiro–Wilk test, and Welch’s t-test was used in cases of heteroscedasticity.

Supplementary figures



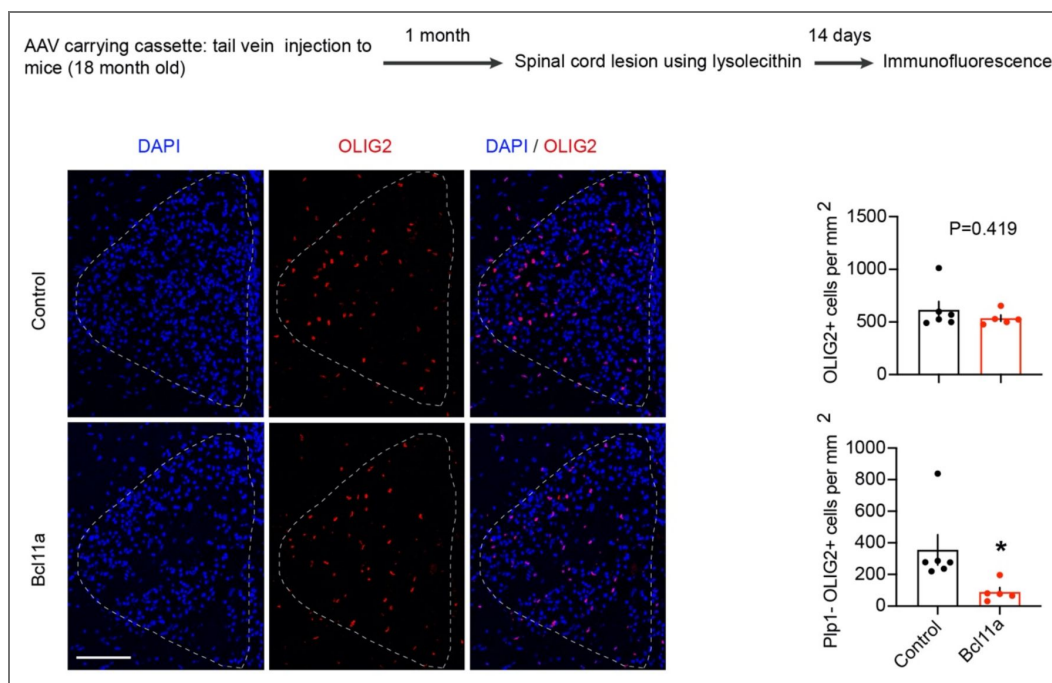
Supplementary Figure 1. Schematic representation of the logic of the gSWITCH web application (for detail see 'Materials and Methods' section). The gSWITCH application is accessible at <https://altoslabs.shinyapps.io/gSWITCH/>. Red circles indicate the start and the end state of a series (time or lineage stages or any other variable dependent series). GLM fit and hypothesis testing followed by contrast

analysis between start and the end state, generates a subset of genes on which we apply maximum (max) or minimum (min) criteria within a closed interval while taking account of monotonicity trend as well as further specific contrasts between groups. This allows identification of four types of switches. Detail criteria are shown inside the boxes and the trend pattern are represented by coloured dashed-straight lines. Arc or solid curve lines indicate statistical significance for specific contrast. Adj p value: Adjusted p value [Benjamini–Hochberg FDR after LRT or Benjamini–Hochberg FDR after Wald test for specific contrast analysis]. u.d.: user defined value. Note that to run gSWITCH, RNA-seq data are required (bulk or scRNA-seq flattened to pseudo bulk), where the number of series or time points or group ≥ 3 and the replicates per time point or group ≥ 2 . gSWITCH also identifies transcription factors (TFs) within the switches and provides a summary of the total number of switches and TFs per switch type.



Supplementary Figure 2. Fidelity of protein expression from in vitro-transcribed Bcl11a mRNA; mRNA transfection efficiency in aged OPCs; and fidelity of AAV construct expression in the OL lineage and infection efficiency in aged mouse brain.

A) Western blot analysis using antibodies to BCL11A and β -actin (ACTB), with ACTB serving as a loading control, in MCF7 cells transfected with Bcl11a mRNA or Gfp mRNA (control). **B)** Immunofluorescence analysis using antibodies to OLIG2 and GFP in aged OPCs transfected with Gfp mRNA. Quantification of GFP+OLIG2+ cells were plotted as percentage of OLIG2+ cells. $n = 3$ independent experiments (each using OPCs isolated from pooled brains of 2 aged rats). mean $\pm$ SEM. Scale bar: 22 μ m. **C)** AAV carrying constructs are shown in Figure 6A. Left: Immunofluorescence analysis of OLIG2 and GFP immunostaining. Right top: Quantification of OLIG2+GFP+ cells were plotted as a percentage of GFP+DAPI+ cells. Right bottom: quantification of OLIG2+GFP+ cells were plotted as a percentage to OLIG2+ cells. $n = 3$ mice, mean+SEM. P: p-value, Student's t test (unpaired, two tailed). Scale bar: 20 μ m.



Supplementary Figure 3. Immunofluorescence analysis using antibody to OLIG2 after OL lineage specific Bcl11a expression in aged mice.

The AAV containing the Sox10-driven construct carrying either Gfp alone (control) or Bcl11a linked to Gfp via a self-cleaving peptide element (P2A) was delivered by tail-vein injection (see Figure 6A). In the lesioned area, the number of OLIG2+ cells per mm² was counted and plotted. In the same region, the number of Plp1⁻ OLIG2+ cells per mm² was also calculated and plotted (see Figure 6B for Plp1⁺ cells per mm²). n=5-6 mice, mean+SEM, P: p-value, ** P<0.01, Student's t test (unpaired, two tailed). scale bar: 100 μ m. The white dotted line demarcates the lesion area.

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Additional information

Data availability

RNA-seq data have been deposited in GEO under accession numbers GSE301578 and GSE303317. The data will be made publicly available upon posting of the reviewed preprint.

Author contributions

Conceptualization: T.G. and R.J.M.F.; formal analysis: T.G.; developer of gSWITCH application: T.G., Computational molecular evolution analysis: T.G. and N.G.; investigation: T.G., R.B., C.Z., AS; methodology: T.G., R.B., C.Z., AS, N.G., R.J.M.F.; project administration: T.G. and R.J.M.F.; supervision: R.J.M.F. and T.G.; writing – original draft, T.G. and R.J.M.F.; writing – review & editing, T.G., R.J.M.F., N.G., R.B., C.Z. and A.S.

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Additional files

Supplementary Table 1. All switches during OL-lineage progression [↗](#)

Supplementary Table 2. Transcription factor (TF) switches during OL-lineage progression [↗](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

This manuscript by Ghosh and colleagues investigates the transcriptional changes within the oligodendrocyte lineage that contribute to age-related declines in oligodendrocyte differentiation and myelination. Combining bulk RNA-Seq on acutely purified oligodendrocyte lineage cells with bioinformatic approaches, the authors identify groups of genes that show different patterns of dynamic regulation during differentiation (which they term "switch" genes, or "switches"). A subset of these switch genes is differentially regulated with age. The authors identify two transcription factors, Bcl11a and Foxm1, that are downregulated during differentiation, have predicted binding site enrichment at other switch genes, and are downregulated in aged OPCs. Functionally testing Bcl11a, the authors show that Bcl11a knockdown inhibits the differentiation of young OPCs in culture, whereas overexpression promotes the differentiation of aged OPCs. Viral expression of Bcl11a in Sox10-expressing cells accelerates the formation of Plp1+ oligodendrocytes in aged rodents following lyssolecithin induced demyelination.

Strengths:

The work is clearly presented and addresses an important biological problem. The bioinformatic approaches used in the manuscript are powerful, and the identification of Bcl11a as a modulator of oligodendrocyte differentiation is a novel finding. The combined in vitro and in vivo approaches to assess the function of Bcl11a in oligodendrocyte differentiation are a substantial strength of the work.

Weaknesses:

Although the PCA plots show distinct and reproducible global gene expression differences between the different isolated cell populations, the authors do not present a figure showing expression levels of typical stage-specific markers (e.g., Pdgfra, Pcdh15, C1ql1 for OPCs, Bcas1, Enpp6, Gpr17 for preOLs, Mobp, Mog, etc. for OLs) or confirm the absence of markers of other lineages (astrocytes, neurons, microglia, etc.). This makes it difficult to evaluate the success of their cell isolation strategy at different ages without reanalyzing the raw data. In addition, other publicly available datasets (e.g., the Barres lab bulk RNA-Seq datasets from PMID 25186741 or the Castelo-Branco lab single cell datasets from PMID 27284195) do not show downregulation of Bcl11a during OL differentiation as is described here - this apparent discrepancy is not discussed.

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Reviewer #2 (Public review):

Aging poses a significant challenge to the regenerative capacity of oligodendrocyte precursor cells (OPCs) to differentiate and myelinate neuronal axons. Myelin abnormalities accumulate with age, and it is likely that the ability of OPCs to differentiate into myelinating oligodendrocytes becomes progressively impaired during aging, leading to inefficient turnover of damaged myelin and oligodendrocytes, as well as reduced adaptive myelination. Understanding the molecular mechanisms underlying the compromised capacity of aged OPCs is therefore critical for addressing age-related white matter decline.

This study aims to decipher the intrinsic molecular changes that occur in aged OPCs. By profiling differentially expressed transcription factors (TFs) between young and aged OPCs, and by employing a novel bioinformatic tool to identify key TFs that undergo dynamic changes across distinct stages of OPC differentiation, the authors identify Bcl11a as a potential regulator. Bcl11a is highly expressed in young OPCs but markedly reduced in aged cells. Functional experiments further demonstrate that while Bcl11a does not affect OPC proliferation, it significantly promotes the differentiation of aged OPCs. Importantly, this effect is also observed *in vivo* following demyelinating injury in aged mice.

While the study provides compelling evidence that BCL11A represents a limiting factor for OPC differentiation during ageing, the downstream targets and molecular mechanisms through which BCL11A exerts its effects are not directly addressed. As such, the work should be interpreted primarily as identifying a key regulatory node rather than a fully defined molecular pathway.

Overall, this study offers valuable insights into the age-related loss of regenerative capacity in the central nervous system and introduces a computational framework that may be broadly useful for investigating dynamic gene regulation in other biological contexts.

Major Points:

- (1) MACS mouse anti-A2B5 microbeads are not OPC-specific and may also label astrocyte precursor cells or immature astrocytes. How do the authors justify this caveat? Could some of the claimed "OPC-specific" switch genes in fact be enriched in astrocyte lineage cells?
- (2) Overall, Figures 1 and 2 are not very informative in terms of biological insight. The authors should provide more detail in the main figures regarding the enriched gene sets associated with each of the Type 1-4 switch categories. For example, summarizing the top Gene Ontology terms for each switch type would greatly enhance interpretability.
- (3) A similar issue applies to Figure 3. The authors should explicitly specify the transcription factors in the main figure, particularly the 27 TFs identified through the ENCODE/ReMap2 analysis.
- (4) Have the authors validated Bcl11a expression across different CNS cell types and between young and aged conditions using independent methods such as qPCR, immunofluorescence, or western blotting?
- (5) Regarding OPC aging, an open question is whether the reduced differentiation capacity of aged OPCs is an intrinsic property of the cells themselves or whether it results from prolonged exposure to an aging environment that induces non-cell-autonomous epigenetic or genetic changes, thereby rendering OPCs less efficient at differentiating. It would be helpful if the authors could expand on this point in the Discussion, with reference to relevant previous studies and experimental evidence.

- (6) Do the authors observe a change in the number or density of OPCs between young and aged mice?
- (7) The in vivo characterization of Bcl11a overexpression using the AAV-based approach appears incomplete. Do aged mice overexpressing Bcl11a in Sox10⁺ cells exhibit reduced age-related myelin degeneration under baseline conditions? In the LPC model, do the authors observe differences in lesion size and/or remyelination efficiency?
- (8) Are the authors presenting gSWITCH for the first time in this manuscript? Given that the gSWITCH framework is novel and central to the study, its conceptual contribution could be emphasized more strongly. A brief comparison with existing trajectory- or pattern-based methods-ideally in the main text around Figure 1-would help readers better appreciate its novelty.
- (9) The evolutionary analysis also appears somewhat disconnected from the rest of the study. Could the authors leverage available public datasets to test whether a similar Bcl11a expression trajectory is observed in human oligodendrocyte lineage cells?

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